



CJC4344

Stereo DAC with 24-Bit, 192 kHz Stereo D/A Converter

Preliminary

Edition	Author	Date	Description
V1.0	By ZN	2015.10	The first draft

Preliminary

DESCRIPTION

The CJC4344 is a high quality stereo DAC designed for portable multimedia applications. The CJC4344 is a complete, stereo digital-to-analog output systems including interpolation, multi-bit D/A conversion and output analog filtering in a 10-pin package.

The CJC4344 support all major audio data interface formats, and the individual devices differ only in the supported interface format. The device including digital interpolation, third-order multi-bit delta-sigma digital-to-analog conversion, digital de-emphasis, analog filtering. the CJC4344 is available in a 10-pin MSOP package in Commercial (-40°C to +85°C) grade.

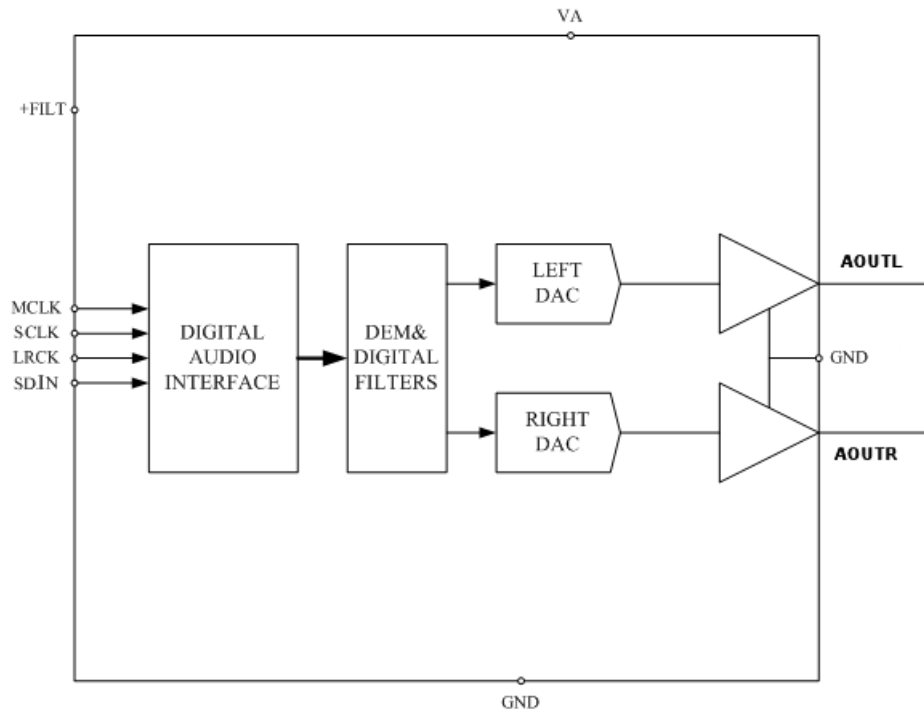
FEATURES

- DAC SNR 96dB ('A' weighted)
- THD -82dB
- Architecture
 - Filtered Line-level Outputs
 - 2/3VRMS Full-scale Output
- VCC/2 common mode
- Supports sample rates up to 192 kHz
- 24-bit I²S input
- Msop10 pin package

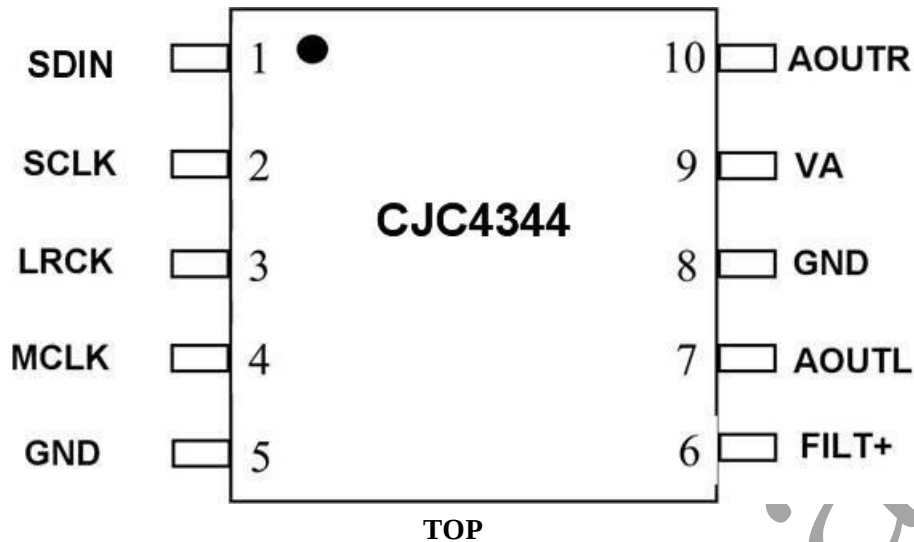
APPLICATIONS

- Handheld gaming
- Mobile multimedia
- DVD players
- set-top boxes
- digital TVs

BLOCK DIAGRAM



PIN CONFIGURATION



PIN DESCRIPTION

PIN NO	NAME	I/O	DESCRIPTION
1	SDIN	I	Serial Audio Data Input - Input for two's complement serial audio data.
2	SCLK	I	Serial Clock - Serial clock for the serial audio interface.
3	LRCK	I	Left / Right Clock - Determines which channel, Left or Right, is currently active on the serial audio data line.
4	MCLK	I	Master Clock - Clock source for the delta-sigma modulator and digital filters.
5	GND	-	Ground - Ground reference.
6	FILT+	O	Positive Voltage Reference - Positive reference voltage for the internal sampling circuits.
7	AOUTL		Analog L Outputs - The full-scale analog line output level is specified in the Analog Characteris.
8	GND	-	Ground - Ground reference.
9	VA	O	Analog Power - Positive power supply for the analog and regulator for the digital core logic sections.
10	AOUTR		Analog R Outputs - The full-scale analog line output level is specified in the Analog Characteris.

CHARACTERISTICS AND SPECIFICATIONS

GND = 0 V; all voltages with respect to ground.(Note1)

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
DC power supply	VA	3.3	5.0	5.5	V
Interface power	VL				V

Ambient operating temperature (power applied) -CSZ	TA	-40	27	85	°C
---	----	-----	----	----	----

Note 1: Device functional operation is guaranteed within these limits. Functionality is not guaranteed or implied outside of these limits. Operation outside of these limits may adversely affect device reliability.

ABSOLUTE MAXIMUM RATINGS

GND = 0 V; all voltages with respect to ground.

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
DC power supply Low voltage	VA	-0.3		6	V
analog power	VL	-0.3		6	V
Interface power	VL	-0.3		6	V
Input current, any pin except supplies	I _{in}	-		±10	mA
Digital input voltage (Note2) Digital interface	V _{IN-L}	-0.3		VL+0.4	V

Normal operation is not guaranteed at these extremes.

Note 2: The maximum over/under voltage is limited by the input current except on the power supply pin.

ELECTRICAL CHARACTERISTICS

Test Conditions

VA=5V, VL=3.3V, GND=0V, TA=+25°C, fs=48kHz, MCLK=256fs, 24-bit data, unless otherwise stated.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNIT
Analogue Output						
Output Level		0dBFS	1.89	2.1	2.31	V _{rms}
Master Clock Frequency	fCLK		7.6		55.3	MHz
Duty Cycle	dCLK		45		55	%
Input Sample Rate		Fs	30		216	KHz
All MCLK/LRCK ratios combined						
Load Impedance				3		KΩ
Load Capacitance				100		pF
DAC to Line-Out						
Signal to Noise Ratio	SNR	RL = 10kΩ A-weighted	98	100	102	dB
		RL = 10kΩ Un-weighted	96	98	100	dB
Total Harmonic Distortion Plus Noise	THD+N	RL = 10kΩ	83	85	86	dB
Channel Separation		1kHz	90	95	100	dB

Power Supply Rejection Ratio	PSRR	1kHz	55	60	65	dB
Digital Logic Levels						
Input HIGH Level	V _{IH}		0.9 DBVDD			V
Input LOW Level	V _{IL}				0.3 DBVDD	V
Output HIGH Level	V _{OH}	I _{OH} = 1mA	0.9 DBVDD			V
Output LOW Level	V _{OL}	I _{OL} = -1mA			0.1 DBVDD	V
Input Capacitance						pF
Input Leakage					+/-0.9	A

Filter Characteristics

(1) (T_a=T_{min}~T_{max}; V_A=5V, V_L=3.3V, f_s=48kHz)

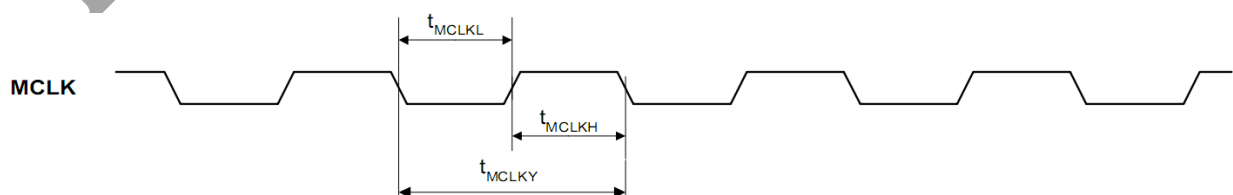
Parameter	Symbol	Min	Typ	Max	Units
DAC Digital Filter:					
Passband (Note3)	±0.01dB	PB	0	20.0	kHz
	-6.0dB	-	22.05	-	kHz
Stopband (Note3)		SB	24.1		kHz
Passband Ripple		PR		±0.01	dB
Stopband Attenuation		SA	65		dB
Group Delay (Note4)		GD	-	24.0	1/fs
Digital Filter + SCF + CTF:					
Frequency	0~20 kHz	FR	-	±0.02	dB
Response	~40 kHz (Note5)	-	±1.0	-	dB

Note 3. The passband and stopband frequencies is proportional to f_s.

Note 4. The delay occurred by digital counter, this time is the time from setting the 16/24bit data of input register to analog signal outputting from both channels .

Note 5. f_s=96kHz.

SYSTEM CLOCK TIMING

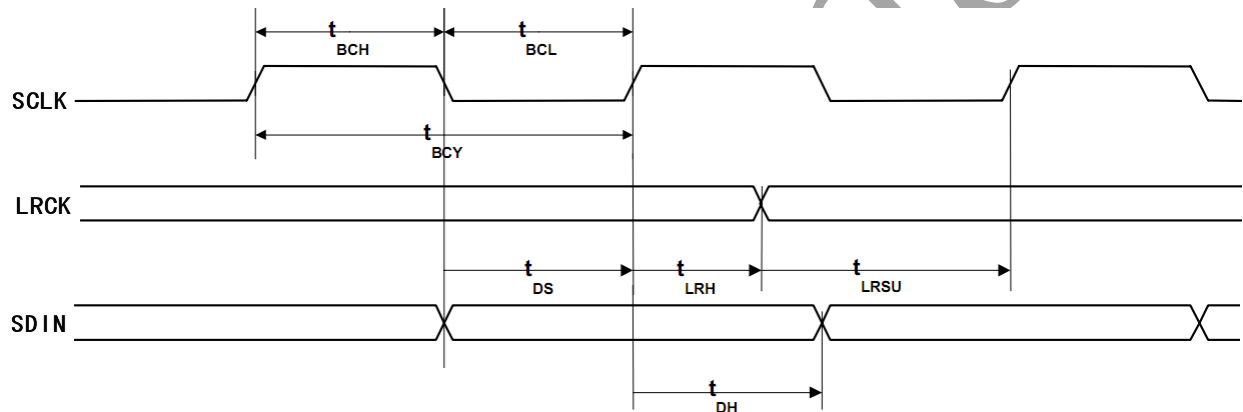


System Clock Timing Requirements

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
System Clock Timing Information					
MCLK System clock pulse width high	T_{MCLKL}	21			ns
MCLK System clock pulse width low	T_{MCLKH}	21			ns
MCLK System clock cycle time	T_{MCLKY}	54			ns
MCLK duty cycle	T_{MCLKDS}	60:40		40:60	ns

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
System Clock Timing Information					
MCLK System clock pulse width high	T_{MCLKL}	10			ns
MCLK System clock pulse width low	T_{MCLKH}	10			ns
MCLK System clock cycle time	T_{MCLKY}	27			ns

AUDIO INTERFACE TIMING

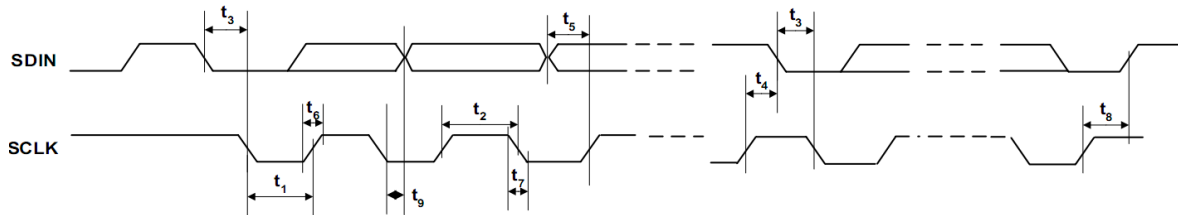


Digital Audio Data Timing

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
Audio Data Input Timing Information					
SCLK cycle time	t_{BCY}	50			ns
SCLK pulse width high	t_{BCH}	20			ns
SCLK pulse width low	t_{BCL}	20			ns
LRCK set-up time to BCLK rising edge	t_{LRSU}	10			ns
LRCK hold time from BCLK rising edge	t_{LRH}	10			ns
SDIN hold time from BCLK rising edge	t_{DH}	10			ns

Note: SCLK period should always be greater than or equal to MCLK period.

CONTROL INTERFACE TIMING – 2-WIRE MODE



Control Interface Timing – 2-Wire Serial Control Mode

Test Conditions

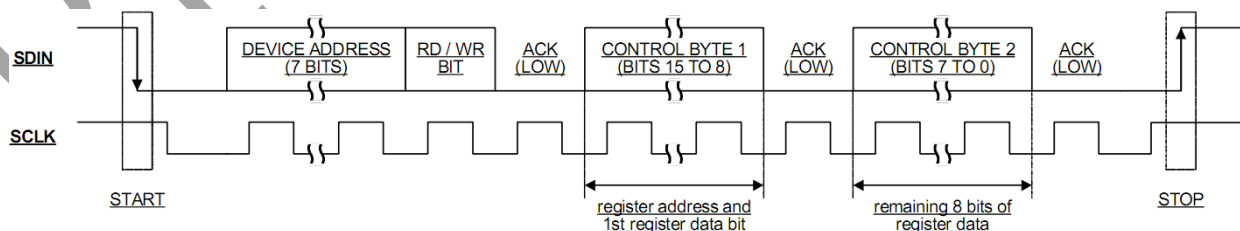
VA = 5V, VL = 3.3V, TA = +25 C, Slave Mode, fs = 48kHz, MCLK = 256fs, 24-bit data, unless otherwise stated.

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
Program Register Input Information					
SCLK Frequency		0		526	KHZ
SCLK Low Pulse-Width	t1	1.3			us
SCLK High Pulse-Width	t2	600			ns
Hold Time (Start Condition)	t3	600			ns
Setup Time (Start Condition)	t4	600			ns
Data Setup Time	t5	100			ns
SDIN, SCLK Rise Time	t6			300	ns
SDIN, SCLK Fall Time	t7			300	ns
Setup Time (Stop Condition)	t8	600			ns
Data Hold Time	t9			900	ns
Pulse width of spikes that will be suppressed	tps	0		5	ns

2-WIRE SERIAL CONTROL MODE

The CJC4344 supports software control via a 2-wire serial bus.

The device address is Write 00110100 (0x34h) and Read 00110101(0x35h)



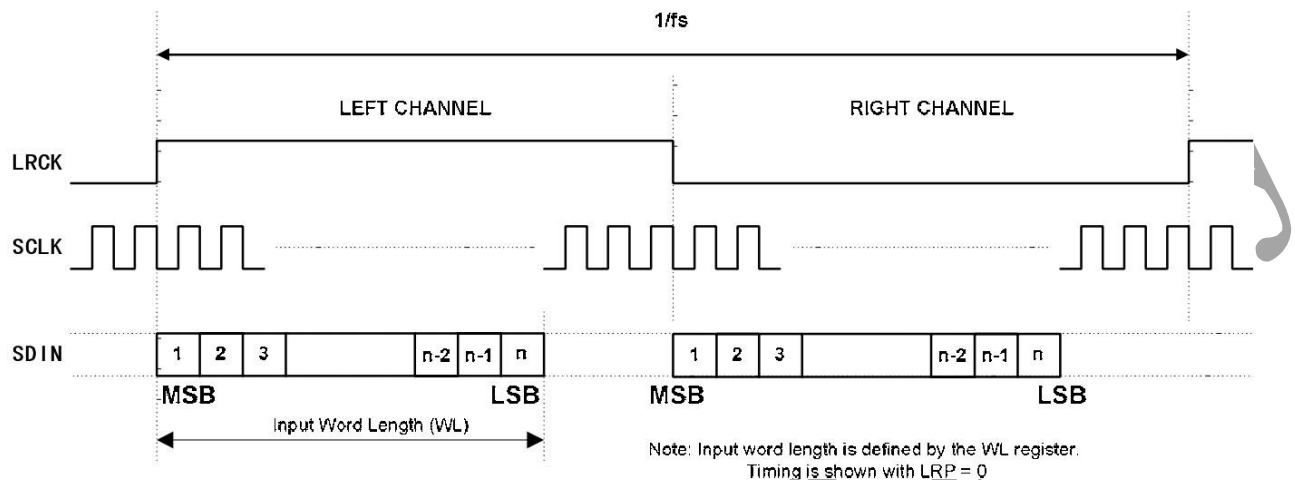
2-Wire Serial Control Interface

Note:

MCLK must be provided before initializing registers.

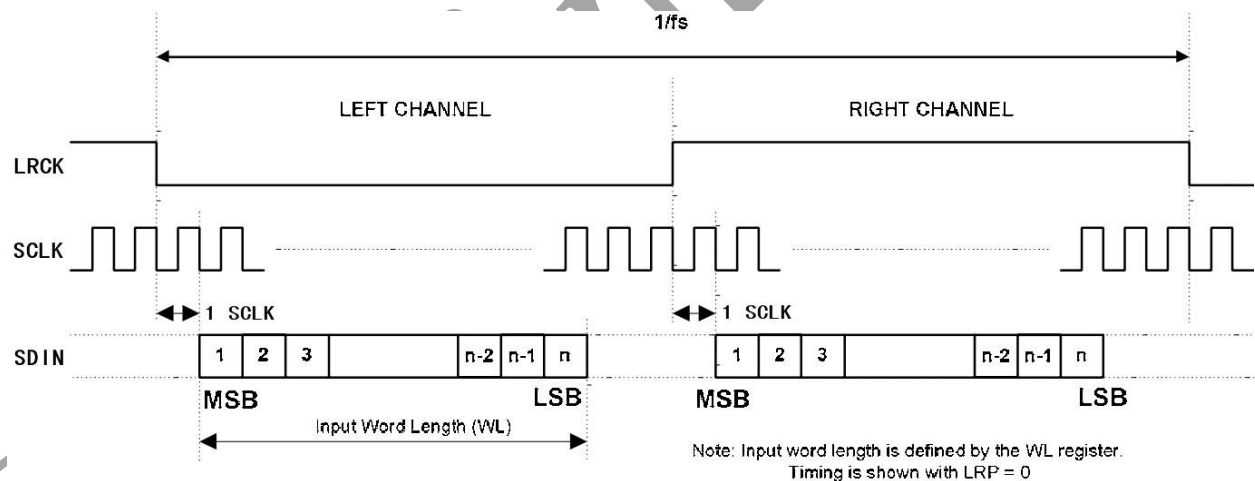
Audio Interface Format

In Left Justified mode, the MSB is available on the first rising edge of SCLK following a LRCLK transition. The other bits up to the LSB are then transmitted in order. Depending on word length, SCLK frequency and sample rate, there may be unused SCLK cycles before each LRCLK transition



Left Justified Audio Interface (assuming n-bit word length)

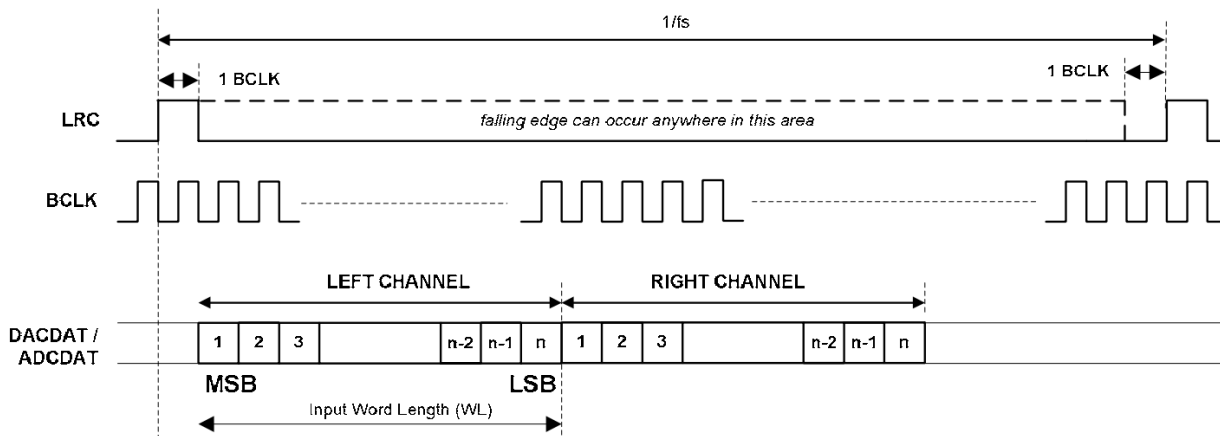
In I2S mode, the MSB is available on the second rising edge of SCLK following a LRCLK transition. The other bits up to the LSB are then transmitted in order. Depending on word length, SCLK frequency and sample rate, there may be unused SCLK cycles between the LSB of one sample and the MSB of the next.



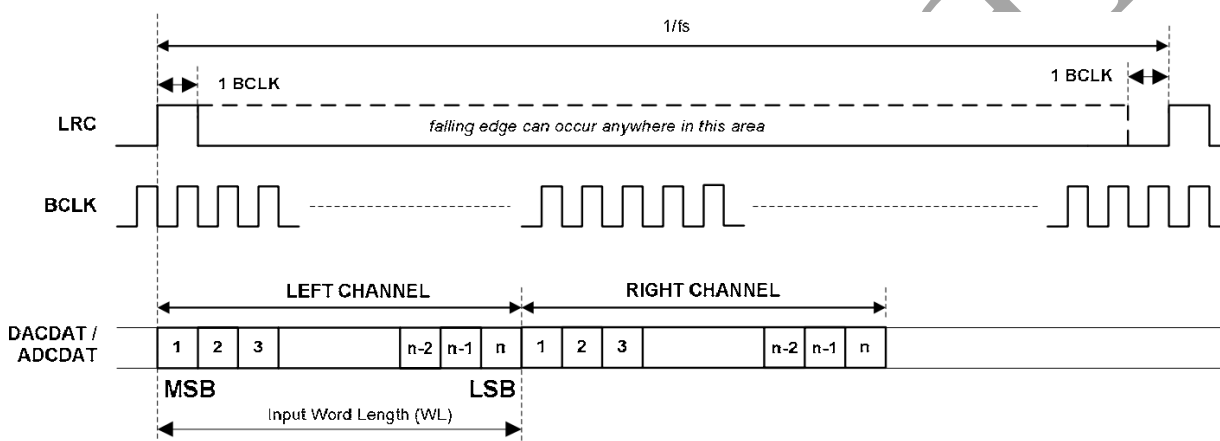
I2S Justified Audio Interface (assuming n-bit word length)

In DSP/PCM mode, the left channel MSB is available on either the 1 (mode B) or 2 (mode A) rising edge of SCLK (selectable by LRP) following a rising edge of LRCLK. Right channel data immediately follows left channel data. Depending on word length, SCLK frequency and sample rate, there may be unused SCLK cycles between the LSB of the right channel data and the next sample. it is possible to use any length of frame pulse

less than $1/f_s$, providing the falling edge of the frame pulse occurs greater than one SCLK period before the rising edge of the next frame pulse.



DSP/PCM Mode Audio Interface (mode A, LRP=0)



DSP/PCM Mode Audio Interface (mode B, LRP=0)

Grounding and Power Supply Decoupling

As with any high-resolution converter, the CJC4344 requires careful attention to power supply and grounding arrangements if its potential performance is to be realized. The “Typical Connection Diagram” on page 23 shows the recommended power arrangements with VA and GND connected to clean supplies.

It is strongly recommended that a single ground plane be used with the GND pins connected to the common plane; this is important because both pin 5 and pin 8 provide analog ground reference to the CJC4344. Should it be necessary to split the ground planes, the CJC4344 should be placed entirely in the analog plane. In this configuration, it is critical that the digital and analog ground planes be tied together with a low-impedance connection, ideally a strip of copper on the printed circuit board, at a single point near the CJC4344.

All signals, especially clocks, should be kept away from the FILT+ pin in order to avoid unwanted coupling into the DAC.

System Clocking

The external clocks, which are required to operate the CJC4344, are MCLK, BICK and LRCK. The master clock (MCLK) should be synchronized with LRCK, but the phase is not critical. The MCLK is used to operate the digital interpolation filter and the Delta-Sigma modulator.

The system clocking have two operate MODE, are AUTO detect MODE and I2C Setting MODE.

The MCLK frequency is detected from the relation between MCLK and LRCK automatically.

AUTO detect MODE

LRCK (kHz)	MCLK (MHz)						
	128x	192x	256x	384x	512x	768x	1024x
32	-	-	8.1920	12.2880	16.3840	24.5760	32.7680
44.1	-	-	11.2896	16.9344	22.5792	33.8688	45.1580
48	-	-	12.2880	18.4320	24.5760	36.8640	49.1520
88.2	11.2896	16.9344	22.5792	33.8688	45.1584	-	
96	12.2880	18.4320	24.5760	36.8640	49.1520	-	
176.4	22.5792	33.8688	45.1584	-	-	-	
192	24.5760	36.8640	49.1520	-	-	-	

De-emphasis Filter

After passing through the graphic equaliser filters, digital 'de-emphasis' can be applied to the audio data if necessary (e.g. when the data comes from a CD with pre-emphasis used in the recording). De-emphasis filtering is available for sample rates of 48kHz, 44.1kHz and 32kHz.

REGISTER ADDRESS	BIT	LABEL	DEFAULT	DESCRIPTION
(05H) ADC and DAC control	2:1	DEEMP [1:0]	00	De-emphasis Control 11 = 48kHz sample rate 10 = 44.1kHz sample rate 01 = 32kHz sample rate 00 = No De-emphasis

When the SCLK/DEM pin is connected to VL (internal SCLK mode), the 44.1 kHz de-emphasis filter is activated. When the SCLK/DEM pin is connected to GND, the de-emphasis filter is disabled.

Each value of SR[5:0] selects one combination of MCLK division ratios and hence one combination of sample rates. Since all sample rates are generated by dividing MCLK, their accuracy depends on the accuracy of MCLK. If MCLK changes, the sample rates change proportionately.

I2C Setting MODE

MCLK CLKDIV2=0 CLKDIV4=0	MCLK CLKDIV2=1	MCLK CLKDIV4=1	SAMPLE RATE	SR [5:0]	BCLK
12.288 MHz	24.576 MHz	49.152MHz	8 kHz (MCLK/1536)	010101	MCLK/24
			12 kHz (MCLK/1024)	011111	MCLK/16
			16 kHz (MCLK/768)	011001	MCLK/12
			24 kHz (MCLK/512)	000011	MCLK/8
			32 kHz (MCLK/384)	010001	MCLK/6
			48 kHz (MCLK/256)	000110	MCLK/4
			96 kHz (MCLK/128)	001010	MCLK/2
			192kHz (MCLK/64)(Note6)	000000	MCLK/2
8.1920 MHz	16.384 MHz	32.768 MHz	8 kHz (MCLK/1024)	011110	MCLK/16
			16 kHz (MCLK/512)	010011	MCLK/8
			32 kHz (MCLK/256)	010111	MCLK/4
			64 kHz (MCLK/128)	011011	MCLK/2
			128kHz (MCLK/128)(Note7)	000001	MCLK/2
11.2896MHz	22.5792MHz	45.1584MHz	8.0182 kHz (MCLK/1408)	001101	MCLK/22
			11.025 kHz (MCLK/1024)	001110	MCLK/16
			22.05 kHz (MCLK/512)	010010	MCLK/8
			44.1 kHz (MCLK/256)	010110	MCLK/4
			88.2 kHz (MCLK/128)	011010	MCLK/2
			176.4 kHz (MCLK/64) (Note8)	010000	MCLK/2
18.432MHz	36.864MHz		8 kHz (MCLK/2304)	110101	MCLK/36
			12 kHz (MCLK/1536)	111110	MCLK/24
			16 kHz (MCLK/1152)	111001	MCLK/18
			24 kHz (MCLK/768)	110011	MCLK/12
			32 kHz (MCLK/576)	110001	MCLK/9
			48 kHz (MCLK/384)	100110	MCLK/6
			96 kHz (MCLK/192)	101010	MCLK/3
			192 kHz (MCLK/96) (Note9)	100000	MCLK/3
16.9344MHz	33.8688MHz		8.0182 kHz (MCLK/2112)	101101	MCLK/33
			11.025 kHz (MCLK/1536)	101110	MCLK/24
			22.05 kHz (MCLK/768)	110010	MCLK/12
			44.1 kHz (MCLK/384)	110111	MCLK/6
			88.2 kHz (MCLK/192)	111011	MCLK/3
			176.4 kHz (MCLK/96) (Note10)	100001	MCLK/3

Note6: Fs=192kHz,MCLK=24.576MHz must be provided CLKDIV2=1 or MCLK=48.152MHz CLKDIV4=1

Note7-10: provided and Note identical

Power-Down Sequence

Follow the power-down sequence below:

1. For minimal pops, set the input digital data (SDIN) to zero for at least 8192 consecutive samples.
2. Remove the MCLK signal without applying any glitched pulses to the MCLK pin.
3. Remove the power supply voltages.

Note: A glitched pulse is any pulse that is shorter than the period defined by the minimum/maximum MCLK signal duty cycle specification and the nominal frequency of the input MCLK signal. A transient may occur on the analog outputs if the MCLK signal duty cycle specification is violated when the MCLK signal is removed during normal operation;

VCC/2 COMMON MODE

The CS4344 integrates complete interfaces to VCC/2 common mode and 2/3 VRMS Ground-centered mode, set register POWER ANAGEMENT2 select. VCC/2 common mode external link ClassD AMP.

REGISTER MAP

ADDR	REGISTER NAME	D7	D6	D5	D4	D3	D2	D1	D0
00H	POWER ANAGEMENT1	VMID	BUFFERCP	COMP CP	VEESEL	-	-	-	-
01H	POWER ANAGEMENT2	-	AOUTA	AOUTB	BOUTB	BOUTB	-	DACL	DACR
02H	DIGITAL AUDIO INTERFACE FORMAT	BCLKINV	-	LRSWAP	LRP	WL1	WL0	FORMAT[1:0]	
03H	DAC CONTROL	-	-	-	-	-	DEEMP[1:0]		-
04H	ADDITIONAL CONTROL	CLOCKAUTO	-	MUTE	-	-	OTD	-	GAINSE
05H	LEFT CHANNEL DIGITAL VOLUME1	LDACVOL[7:0]							
06H	LEFT CHANNEL DIGITAL VOLUME2	-	-	-	-	-	-	-	LDVU
07H	RIGHT CHANNEL DIGITAL VOLUME1	RDACVOL[7:0]							
08H	RIGHT CHANNEL DIGITAL VOLUME2	-	-	-	-	-	-	-	RDVU
09H	CLOCKING AND SAMPLE RATE CONTROL	CLKDIV4	CLKDIV2	SR5	SR4	SR3	SR2	SR1	SR0
0AH	TEST	-	TEST_ANA	TEST_ANA[1:0]		-	-	-	

REGISTER DEFINITIONS

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
00H	Power Management1	VMID	-	COMPCP	-	-	-	-	-
	Default	1	0	1	0	0	0	0	0

VMID: Vmid divider enable
0 = Vmid Disabled
1 = Vmid Enable

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
01H	Power Management2	-	-	-	VOUTR	VOUTL	-	DACL	DACR
	Default	0	0	0	1	1	0	1	1

DACR: DAC Right
0 = DAC Power down
1 = DAC Power up

DACL: DAC Left
0 = DAC Power down
1 = DAC Power up

VOUTR: CTF Enable (VCC/2-centered left Output)
0 = VOUT Disabled
1 = VOUT Enable

VOUTL: CTF Enable (VCC/2-centered right Output)
0 = VOUT Disabled
1 = VOUT Enable

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
02H	Digital Audio Interface Format	BCLKINV	-	LRSWAP	LRP	WL1	WL0	FORMAT1	FORMAT0
	Default	0	0	0	0	1	0	1	0

FORMAT1-0: Audio Data Format Select
11 = DSP Mode
10 = I2S Format
01 = Left justified
00 = reserved (do not use this setting)

WL1-0: Audio Data Word Length

- 11 = reserved
- 10 = 24 bits
- 01 = 20 bits
- 00 = 16 bits

LRP: right, left and I2S modes – LRCLK polarity

- 1 = invert LRCLK polarity
- 0 = normal LRCLK polarity

DSP Mode – mode A/B select

- 1 = MSB is available on 1 BCLK rising edge after LRC rising edge (mode B)
- 0 = MSB is available on 2 BCLK rising edge after LRC rising edge (mode A)

LRSWAP: Left/Right channel swap

- 1 = swap left and right DAC data in audio interface
- 0 = output left and right data as normal

BCLKINV: BCLK invert bit (for master and slave modes)

- 0 = BCLK not inverted
- 1 = BCLK inverted

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
03H	DAC control	-	-	-	-	-	DEEMP1	DEEMP0	-
	Default	0	0	0	0	0	0	0	0

DEEMP1-0: De-emphasis Control

- 11 = 48kHz sample rate
- 10 = 44.1kHz sample rate
- 01 = 32kHz sample rate
- 00 = No De-emphasis

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
04H	Additional Control	CLOCKAUTO	-	MUTE	-	-	OTD	-	GAINSE
	Default	0	0	1	0	0	1	0	0

GAINSE: Out gain select

0 = 2VRMS

1 = 3VRMS

OTD: Thermal Shutdown Enable

0 : Disabled

1 : Enable

MUTE: Analog OUT MUTE

0 = mute (signal active)

1 = no mute

ClockAUTO: SYSTEM Clock discern

0 : Auto discern MCLK(default)

1 : manual SET MCLK

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
05H	Left Channel Digital Volume1	LDACVOL[7:0]							
Default		1	1	1	1	1	1	1	1

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
06H	Left Channel Digital Volume2	-	-	-	-	-	-	-	LDVU
Default		0	0	0	0	0	0	0	0

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
07H	Right Channel Digital Volume1	RDACVOL[7:0]							
Default		1	1	1	1	1	1	1	1

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
08H	Right Channel Digital Volume2	-	-	-	-	-	-	-	RDVU
Default		0	0	0	0	0	0	0	0

LDACVOL7-0: Left DAC Digital Volume Control

0000 0000 = Digital Mute

0000 0001 = -127dB

0000 0010 = -126.5dB

... 0.5dB steps up to
1111 1111 = 0dB

LDVU: Left DAC Volume Update

0 = Store LDACVOL in intermediate latch (no gain change)

1 = Update left and right channel gains (left = LDACVOL, right = intermediate latch)

RDACVOL 7-0: Right DAC Digital Volume Control

0000 0000 = Digital Mute

0000 0001 = -127dB

0000 0010 = -126.5dB

... 0.5dB steps up to

1111 1111 = 0dB

RDVU: Right DAC Volume Update

0 = Store RDACVOL in intermediate latch (no gain change)

1 = Update left and right channel gains (left = RDACVOL, right = intermediate latch)

Addr	Register Name	D7	D6	D5	D4	D3	D2	D1	D0
09H	Clocking and Sample Rate Control	CLKDIV4	CLKDIV2	SR5	SR4	SR3	SR2	SR1	SR0
Default		0	0	0	0	0	1	1	0

SR5-0: Sample Rate Control

CLKDIV2: Master Clock Divide by 2

0: MCLK is not divided

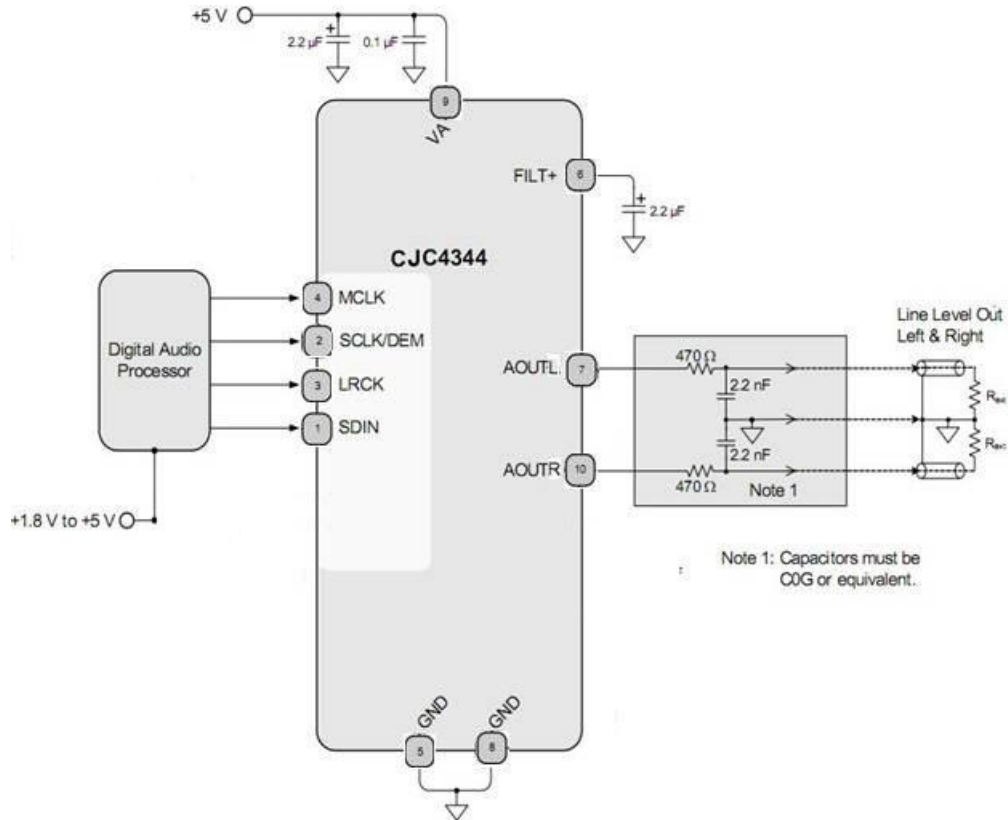
1: MCLK is divided by 2

CLKDIV4: Master Clock Divide by 4

0: MCLK is not divided

1: MCLK is divided by 4

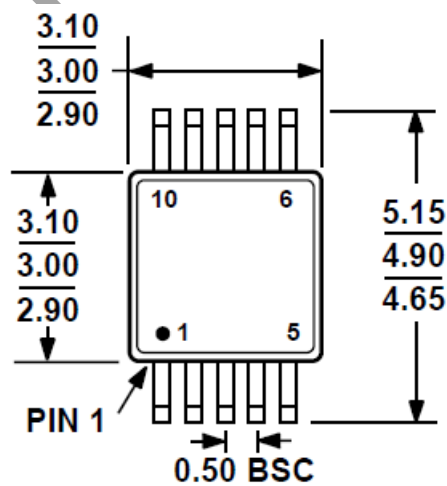
ABSOLUTE MAXIMUM RATINGS System design

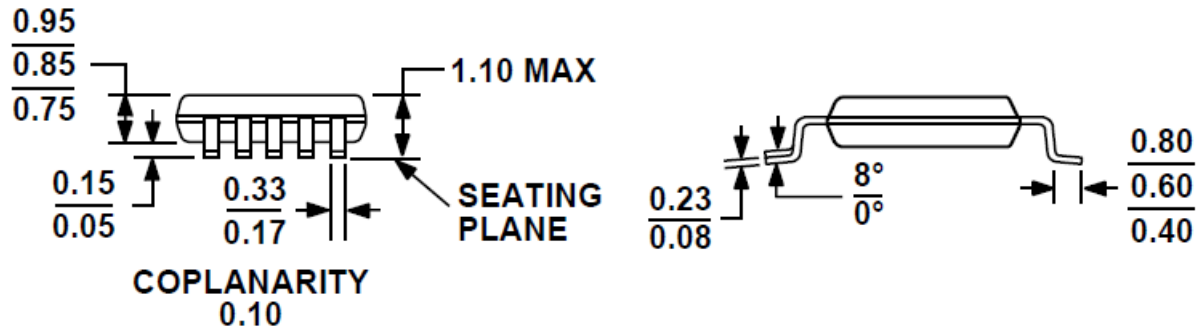


PACKAGE INFORMATION

Dimensions 10-lead Mini Small Outline Package [MSOP]

PACKAGE DRAWING





COMPLIANT TO JEDEC STANDARDS MO-187-BA

		MILLIMETER		
		RS		
DIM	MIN	NOM	MAX	
A	--	--	1.1	
A1	0.05	-	0.15	
A2	0.75	0.85	0.95	
A3	0.3	0.35	0.3	
b	0.19	-	0.23	
b1	0.18	0.2		
c	0.15	-	-	
c1	0.14	0.152		
D	2.9	3		
E	4.7	4.9	0.8	
E1	2.9	3		
e	0.5BSC			
L	0.4		0.7	
L1	0.9BSC			
	0°	-	8°	
71*96				

JEDEC #: MS-012

Controlling Dimension is Millimeters